



Pathways for low carbon Nigeria in 2050 by using NECAL2050

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Due to the adverse impacts of energy production and consumption on the climate, countries across the world are now charged with developing effective strategies that provide energy security and protect the climate. However, studies exploring pathways to reduce energy demand and GHG emissions in Nigeria are limited and have not really focused on what is technically feasible within the country. In this research, we use the Nigerian Energy Calculator 2050 to examine and compare the energy balances and GHG mitigation potentials for four alternative low carbon scenarios which are possible by 2050. Results show that the alternative low carbon pathways will reduce energy consumption, GHG emissions, and air pollution compared to the business-as-usual case. However, our analysis indicates that fossil fuels will still dominate the Nigerian energy system in all scenarios by 2050. Thus, efforts that successfully decouple economic development from GHG emissions are still needed to keep Nigeria on the path of low carbon transition. Our findings further suggest that bioenergy and energy efficiency will play significant roles towards decarbonizing Nigeria. Policy implications for low carbon transition based on our analysis are also discussed. This paper may serve as a reference for developing energy and climate policies in Nigeria.

Introduction

Energy is a vital element for the socio-economic development and well-being of a society [1]. Due to the projected increase in global population, industrialization, and improved living standards, the demand for energy is expected to increase rapidly in the future. However, around 75% of global energy is still produced from fossil fuels such as coal, oil, and natural gas [2]. The combustion of fossil fuels in transportation, electricity generation, and other economic activities releases greenhouse gases (GHGs) into the atmosphere which contributes to global warming and, in turn, leads to climate change [3]. This has become the main reason why energy generation and utilization pattern are given special attention in addressing climate change. The negative impacts of global climate change can be observed in the shrinking of glaciers, breaking up of lakes and river ice, heat waves among others [4].

In Nigeria, the adverse impacts of climate change are also evident. Examples include the shrinking of Lake Chad from about 45,000–3000 km² between 1960 and 2007 [5] and the 2012 flood whose damage amounted to about 1.4% loss of the GDP (US\$ 17 billion) in that year [6]. Owing to the importance of energy in development and the devastating effects of climate change, countries across the world are now charged with developing effective strategies that provide long-term energy security and also protect the climate. In December 2015, the 21st edition of the United Nations Framework Convention on Climate Change (UNFCCC) Conference of Parties (COP-21) was held in Paris. During COP-21, governments across the world reached an agreement to reduce the global average temperature below 2 °C above the level during the pre-industrial era. The Paris Accord was binding for both developed and developing countries. Achieving these ambitious targets requires serious transformation in all sectors of the economy, especially the energy sector, which is the dominant source of GHG emissions globally [7].

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Nigeria has shown commitment to the global efforts towards combating climate change. It joined the UNFCCC in 1994 and ratified its Kyoto Protocol in 2004, and is also a signatory to the Paris Accord. The country submitted its first and second National Communications to the UNFCCC in 2003 and 2014, respectively, and has hosted several Clean Development Mechanism (CDM) projects [5]. However, Nigeria's energy system is still dominated by traditional biomass and fossil fuels which is the major source of GHG emissions in the country. About 50% of Nigerians still depend on solid traditional fuels for cooking while around 60% of the rural dwellers do not have access to electricity [8]. The per capita energy consumption of Nigeria was 763 kg of oil equivalent (kgoe) in 2014, which is only about 40% of the world's average of 1919 kgoe. The human development index (HDI), as of 2015, was low at 0.527 and a global rank of 152 out of 188 countries [9]. Nigeria continues to have a large under-served yet aspiring population, indicating the need for development as a priority for the country. Undoubtedly, energy consumption in Nigeria will continue to increase in the future due to projected population increase and the yearning for socio-economic development. Satisfying unmet energy demand and ensuring rapid economic development without compromising environmental sustainability creates a dilemma for decision makers.

The Nigerian government aims to pursue its developmental goals in a sustainable manner and, thus, has the responsibility to limit its national GHG emissions. Mitigating GHG in Nigeria will require the triangulation of policies, political as well as citizenry will. It requires decisions that will affect the lives of Nigerians in the next decades, such as the use of new technologies and behavioral changes. It will also involve the change of current infrastructures. Hence, decisions need to be made based on a sound understanding of the future challenges in terms of energy demand, supply, and GHG emissions. The year 2050 has been identified as a milestone in the low carbon transition agenda by the Intergovernmental Panel on Climate Change (IPCC), and between 2050–2100, negative GHG emissions will be needed to ensure that the global temperature is kept below 2 °C [10]. Thus, exploring energy pathways to 2050 for Nigeria will help us understand the challenges and opportunities facing the nation in the context of energy security and climate mitigation.

In this research, we examine and compare four alternative scenarios for low carbon development in Nigeria in addition to the reference scenario. To achieve this, the Nigeria Energy Calculator 2050 (NECAL2050) was used to simulate final energy demand and supply, sectoral GHG emissions, per capita GHG emissions, air pollutants as well as policy implications of different scenarios. We also conducted a sensitivity analysis of final energy demand with the household sector of Nigeria as it is the largest energy-consuming sector in the country. It is worthwhile to mention that this study does not recommend a particular scenario for Nigeria. It does, however, show the plausible changes that is required if the energy system evolves through a particular scenario. We believe that this paper will meaningfully contribute to the low carbon development policies of Nigeria and engage key stakeholders in the country on the debate of sustainable energy transition.

The remainder of this paper is structured as follows. Section “Brief literature review” presents a brief review of literature on energy system models and low carbon development studies.

Section “Methodology and assumptions” explains the methodology used based on NECAL2050 and how the scenarios are developed. Section “Results” presents the results of the study in terms of energy demand and supply as well as GHG emissions. Section “Discussion and policy implications” discusses the implications of the results and conclusions are drawn in Section “Conclusions”.

Brief literature review

The concept of low carbon development refers to an economy that is based on low carbon energy sources or has minimal GHG emissions output while ensuring economic growth at the same time. The concept became popular as some developed countries realized that the over-exploitation of the earth resources and consumerism would potentially jeopardize the future of the earth [11]. Energy system models are used as instruments to explore future energy systems with the aim of informing low carbon development policies. Energy system models are broadly classified into two, which are (i) engineering bottom-up models and (ii) macroeconomic top-down models [12]. Bottom-up models are used to explore future energy supply-demand with detailed representation of technologies. They normally do not consider the economic implications of energy/climate policies. Bottom-up models are further classified into the following: simulation (example: LEAP and MAED); optimization (example: MARKAL/TIMES and MESSAGE); and multi-agent models [13–15]. In contrast, the top-down models represent the aggregate economy and are used to evaluate the aggregated impacts of energy/climate policies in terms of monetary value. They do not offer a detailed representation of technologies. They are further grouped into the following: econometric; input-output; computable general equilibrium model (example: GEM-E3); and system dynamics (example: SUS-CLIME) [13,16]. For a detailed review of energy system models, see Refs. [17] and [18].

Over the years, some of these models have been used to explore future energy systems of different regions/countries across the world, and their results have been used to underpin many energy and climate policy decisions [19]. The International Energy Association (IEA) has modeled global GHG mitigation pathways in its energy technology perspective studies [20]. TIAM-WORLD model has been employed in scenario assessment for GHG mitigation [21]. TIAM-WORLD model has also been used to examine the future role of nuclear energy [22], and to analyze the role of renewables and carbon capture and storage (CCS) [23]. In Ireland, Chiodi et al. [24] used the Irish TIMES model to x-ray the energy system implications of the European 2050 climate policies on Ireland's energy system. Elizondo et al. [25], have used the Mexico 2050 calculator to analyze and compare four different carbon mitigation scenarios for Mexico by 2050. Other detailed assessments on GHG emission mitigation using energy system models in different regions across the globe are highlighted in Refs. [26] and [27]. However, studies exploring GHG mitigation pathways for Nigeria are limited and have not really focussed on what is technically feasible within the country in the near future. To our knowledge, the only academic analysis that has explored GHG emissions mitigation scenarios for Nigeria is the work of Emodi et al. [28]. The authors used the LEAP modeling framework to examine future energy demand, energy supply, and corresponding GHG emissions scenarios for Nigeria from 2010 to 2040.

Therefore, there is need for a study that explores low carbon futures that are practically achievable in Nigeria by 2050.

Methodology and assumptions

We used the NECAL2050 model to perform a comparative assessment of energy balances and GHG mitigation potentials for different low carbon scenarios. National 2050 calculator was initially developed by the United Kingdom [29]; after which many other countries across the world followed suit. The documentation of different national calculators can be found in Ref. [30]. NECAL2050 was developed through the memorandum of understanding signed between the United Kingdom Department of Energy and Climate Change (UK-DECC) and the Energy Commission of Nigeria (ECN) in November 2013, at the British High Commission in Abuja, Nigeria. The ECN is the Nigerian governmental agency saddled with the responsibility of strategic planning and coordination of all national energy policies. The NECAL2050 was released in 2015 and was built based on the modeling framework of the UK 2050 pathways calculator.

NECAL2050 is a transparent and open source model of energy, GHG emissions (CO_2 , CH_4 , N_2O , HFCs, PFCs, SF_6), and land use. The model also accounts for air pollutant emissions such as PM_{10} , NO_x , SO_2 , and NMVOC (Non-methane volatile organic compounds). Two versions of the model were developed: the MS Excel-based type and the Web-based type. In this paper, the MS Excel version was employed. NECAL2050 uses an accounting framework algorithm to constantly match energy supply to demand. It uses a set of existing and future technologies to ensure that energy supply satisfies the varying demands. The Excel version used in this paper balances energy demand with supply algorithmically on intervals of five years up to 2050. As an integrated model, NECAL2050 aims to identify secured pathways for energy supply and demand at minimal GHG emissions level from 2010 to 2050.

The demand side of NECAL2050 is split into four sectors; (i) commercial (ii) industry (iii) residential and (iv) transport sector. The demand is determined by the user out of four choices for each demand sector. Then the user also determines how the energy supply would be and then NECAL2050 compares the demand demand-supply figures based on the user choices. The primary energy supply side consists of coal, gas, and oil. When the supply is higher than demand, it is assumed that the domestic energy resources are sufficient to match the demand. However, NECAL2050 depends on imported fossil fuels when domestic energy resources are insufficient to match the demand. Electricity production sources in NECAL2050 include bioenergy, coal, hydro, nuclear, solar, and wind. However, in the case of a negative electricity balance, then electricity is imported to satisfy demand.

Supply of biomass is from agriculture and waste recycling while bioenergy is imported in case of a shortfall in domestic supply. Other secondary energies encompass refinery products and bio-fuels. The aforementioned methodology can be mathematically summarized as follows:

Definition:

$$d_i = \text{sum of sectoral energy demand in year } i \quad (1)$$

$$S_i = \text{sum of energy supplies by energy form in year } i \quad (2)$$

Supply minus losses is:

$$x_i = S_i - l_i \quad (3)$$

Where x_i represents the supply losses and l_i the sum of energy conversion and distribution losses as well as energy use by energy processes in year i

Then:

$$b_i = d_i - x_i \quad (4)$$

Where b_i represents the energy supply and demand difference that has to be balanced either by import or by self-generation in the case of electricity.

It is important to note that there are some uncertainties in the calculator model. GHG emission sources in NECAL2050 are grouped into fuel combustion, industrial processes, solvent, and other product use, agriculture, land use land-use change and forestry (LULUCF), waste, other, bioenergy credit, carbon capture, and storage. However, GHG emissions from major sub-sectors such as electricity production, gas flaring, and transportation are aggregated under emissions from fuel combustion. Thus, deeper analysis on the approaches to reduce emissions from these key sources cannot be achieved using NECAL2050. For every sector in NECAL2050, a list of four different trajectories describing the potential changes that might occur in that sector in the future is given. From the supply side, the trajectories describe the scale of energy production facilities while the demand side describes the technological and behavioral changes/choices that might occur up to 2050. These trajectories are represented as 'levels' in the NECAL2050. Figure 1 shows the scale of efforts in each level towards reducing energy demand and GHG emissions.

Level 1 suggests that no efforts will be put towards low carbon development while already existing policies and infrastructures will grow in similar trend till 2050. Level 2 describes a minimum effort towards low carbon development which is in line with already existing policies and has been described by most experts as the attainable lever. Level 3 describes a greater effort towards decarbonization which will require significant technological changes, behavioral changes as well as huge financial investment.

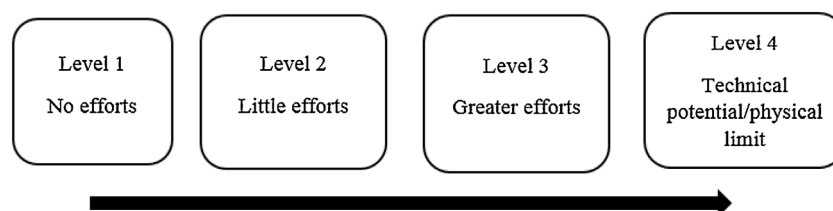


FIGURE 1

Trajectories setting.

TABLE 1

Macroeconomic and demographic assumptions of NECAL2050. Source: [33].

Year	GDP (billion NGN)	Annual GDP growth rate (%)	GDP per capita (NGN)	Population (million persons)	Household size	Number of households (million)
2010	286.28	6.34	1,975.34	159	5.2	30
2020	551.43	6.85	3,010.90	207	5.4	38
2030	1097.47	7.25	4,681.31	265	5.0	53
2040	2209.86	7.25	7,363.80	331	5.0	66
2050	4449.78	7.00	12,163.91	403	5.0	80

Level 4 corresponds to the physical limit for decarbonization of the country. The level doesn't suggest a zero-carbon energy system but an energy system (including conventional and non-convention technologies) that can be technically achieved in Nigeria by 2050. The level is a very ambitious and challenging one and is doubtful considering the current energy situation in the country.

The macroeconomic and demographic assumptions used in the development of NECAL2050 are shown in Table 1. Gross domestic product is projected to grow at an average annual rate of 7%, from 286 billion Nigerian Naira (NGN) in 2010 to around 4450 billion NGN by 2050. The assumptions for the GDP projection are based on the 7.09% average GDP growth rate observed within 2007–2011. Moreover, an earlier study conducted by the ECN for energy demand and supply strategies in the country used 7% GDP growth rate per annum for the reference scenario [31]. Population is expected to increase at an annual rate of 2.75% from 159 million in 2010 to around 403 million by 2050 while the number of households is projected to grow from around 30 million in 2010 to 80 million in 2050, as computed from the National Bureau of Statistics (NBS) 2010 Annual Abstract of Statistics Report [32].

The main reason for selecting the NECAL2050 model for this study is that it was developed locally. Nigeria-specific data and the country's situation on economic and technological options were used in developing the model's trajectories. The model was developed based on vast literature review and consultations with over one hundred local experts, to ensure that the most accurate assumptions are made and the data used are of high quality. Hence, these features give the model a comparative advantage over other models as the results produced by its scenarios actually reflect what is technically/practically attainable in the country within the given timeframe. A detailed description of the model structure is presented in the Appendix A, while the complete model (NECAL2050) is also provided as a Supplementary file to this paper (see Appendix B). The model encompasses all the equations, assumptions, and interconnections of variables used in creating the integrated assessment model. Thus, all of the scenarios and analysis presented here are replicable. Detailed documentation and instructions for using NECAL2050 are available in Ref. [33]. The structure of NECAL2050 is shown in Figure 2.

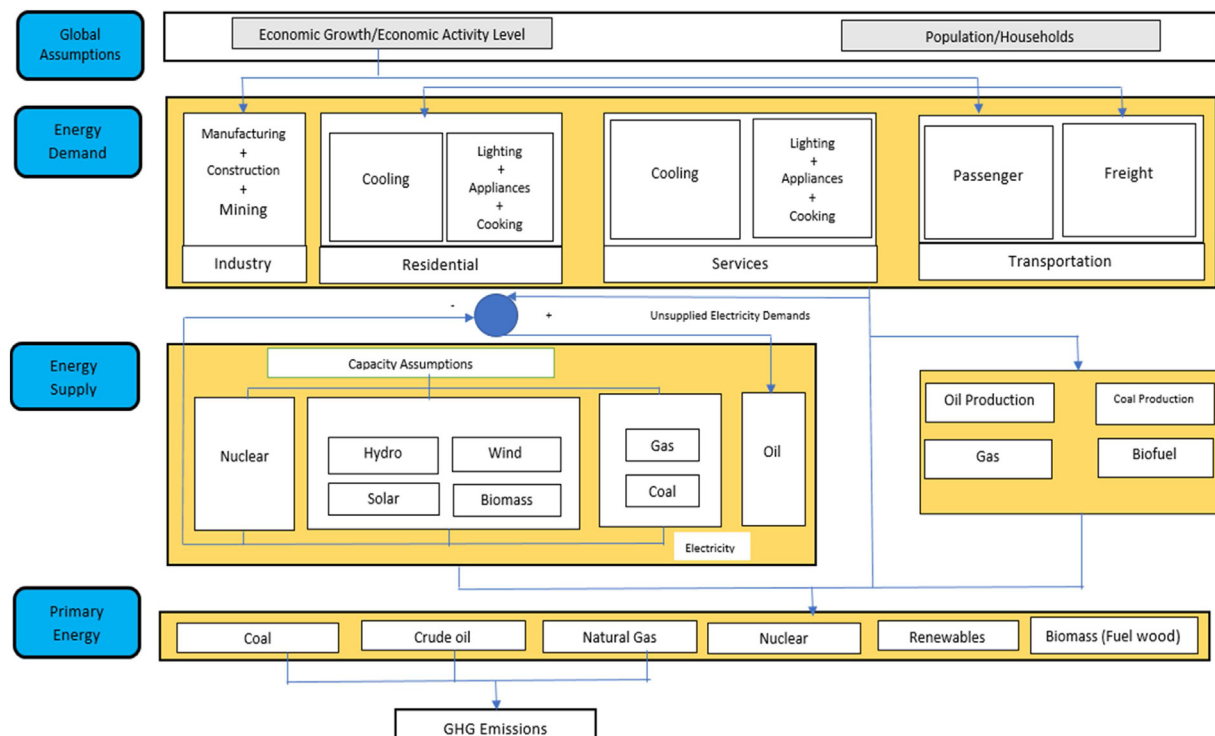


FIGURE 2

Structure of NECAL2050 model.

Scenarios development

Moving towards a low carbon future in Nigeria will entail the deployment of clean energy technologies in all sectors of the country as well as a sustainable lifestyle by the citizens. Here we analyzed the current policies/studies for low carbon development in the country which include the National Renewable Energy & Energy Efficiency Policy (NREEEP) [34], Nigeria Nationally Determined Contribution (NDC) [5], and the World Bank study for low carbon development in Nigeria [35]. A detailed evaluation of these documents shows the various measures which the government wants to and could embrace for low carbon development in the country. The measures can be summarized as follows:

- Large-scale deployment of renewable energy
- Deployment of nuclear energy
- Economy-wide energy efficiency and conservation
- Behavioral change for low carbon lifestyle

To develop scenarios for low carbon transition, these measures were combined to depict the efforts towards each. By changing the

levers of NECAL2050, numerous scenarios which consider what is technically achievable in the country can be generated. We modeled the level 1 trajectory for all sectors of the economy as the reference scenario (S1). The S1 refers to a future where government's policy or intervention is little in terms of improving energy supply, reducing energy demand and mitigating GHG emissions up to 2050. This scenario provides the framework for comparing the energy demand reduction and GHG mitigation potentials of other scenarios. Four additional scenarios are created which are: high renewables + high energy efficiency + behavior (S2); high nuclear + moderate renewables + low energy efficiency (S3); high bioenergy + high energy efficiency + moderate renewables (S4); and shared efforts (S5). The scenarios are described in the next sections. High, moderate, and low classifications were used to describe the extent of penetration of each mitigation measures in the low carbon scenarios. To achieve this in NECAL2050, we selected the lever trajectories that best describe the mitigation measures highlighted in the different scenarios [25]. The levels selected for each scenario are shown in Table 2.

TABLE 2

Levels selected for each scenario.

Sectors	Levers	Levels selected for each scenario				
		S1	S2	S3	S4	S5
Supply						
Electricity generation	Natural gas power stations	1	1	1	1	2
	Biomass power	1	3	2	2	3
	Coal power stations	1	1	1	1	1
	Nuclear power stations	1	1	4	1	2
	Wind	1	4	2	2	2
	Hydroelectric power stations	1	4	2	2	3
	Small hydroelectric power stations	1	4	2	2	2
	Grid connected solar PV	1	4	3	3	3
	Concentrated solar power	1	4	2	2	2
	Stand alone solar photo voltaic	1	4	3	3	3
	Electricity imports	1	1	1	1	1
Bioenergy supply	Agriculture and land use					
	Land dedicated to bioenergy	1	1	1	4	2
	Livestock and their management	1	1	1	4	2
	Volume of waste and recycling	1	1	1	4	2
	Type of fuels from biomass	1	1	1	4	3
	Bioenergy imports	1	1	1	4	2
Demand						
Transport	Domestic passenger transport					
	Domestic transport behavior	1	2	1	2	3
	Shift to low carbon emission transport technology	1	4	3	3	2
	Fuel switch for internal combustion engine	1	4	1	1	2
	Domestic freight	1	2	1	2	3
Households	Residential cooling					
	Cooling demand	1	2	2	3	2
	Efficiency of cooling system	1	4	1	4	2
	Residential lighting, appliances, and cooking					
	Residential lighting, appliances & cooking	1	2	2	3	3
	Technology pathway (cooking)	1	4	1	4	3
Industry	Industrial processes					
	Growth in industry with GDP	1	2	2	2	2
	Energy intensity of industry	1	4	1	4	3
Services	Service sector cooling					
	Service sector demand for cooling	1	2	2	3	3
	Efficiency of cooling system	1	4	1	4	2
	Service sector lighting, appliances, and cooking					
	Service sector lighting, appliances & cooking	1	2	2	3	2
	Technology pathway (cooking)	1	4	1	4	2

High renewables + high energy efficiency + behavior (S2)

In this scenario, we modeled a very high penetration of renewable energy in the electricity sector, economy-wide energy efficiency and the shift towards a more sustainable lifestyle. We assumed that all forms of renewable energies are fully exploited with the exception of bioenergy. Energy-efficient boilers and other best available technologies are employed in the industries as well as energy-efficient vehicles in the transport sector. All households phase out incandescent bulbs and shift to energy-efficient appliances while the use of public mode of transport and cycling is enhanced.

High nuclear + moderate renewables + low energy efficiency (S3)

This scenario presumes that the full potential of nuclear sources of energy is exploited in the country for electricity generation under the assumption that the required infrastructures for its deployment will be made available in Nigeria. All forms of renewable energy are assumed to have a fair share in the energy system while little effort is made towards energy efficiency across all sectors of the economy.

High bioenergy + high energy efficiency + moderate renewables (S4)

This scenario combines bioenergy, energy efficiency, and other renewables. We consider an energy system in which renewables are deployed at a moderate level and the technical limit for exploiting bioenergy is reached in the country. Energy efficiency measures are also considered to have a high penetration across all the sectors in this scenario.

Shared efforts (S5)

This scenario represents an energy system where efforts towards energy supply, energy demand reduction, and GHG mitigation are distributed across all sectors of the economy. On the supply side, a fair proportion of renewable energy technologies and nuclear energy are deployed in the electricity sector. Good infrastructures that will reduce the travel kilometers in cities are developed while a switch in transport mode is also adopted. Consumers use less energy for heating, lighting and other energy services. The energy intensity of the industry sector reduces as a result of energy-efficient technologies.

While modeling these scenarios in the NECAL 2050, we kept the macroeconomic projections constant across all scenarios. The macroeconomic indicators are expected to have an impact on future energy demand by 2050. The increase in energy demand in the service industry, and freight transport sectors is expected due to the increase in socioeconomic variables (GDP and gross output per sector), while residential and passenger transport demand will rise due to the increase in population and household size. However, there is a caveat here since it is uncertain whether these macroeconomic projections will remain constant in the future.

Co-benefits of low carbon scenarios

Most low carbon development strategies will require additional investment compared to the reference scenario [36]. However, the actual cost may be different when the co-benefits of a particular scenario is considered. Additionally, the co-benefits of a low

carbon pathway will go a long way to determine its feasibility. Thus, for effective adoption of a low carbon development strategy by policymakers as well as civil societies, the co-benefits of such a strategy need to be considered by researchers when analyzing scenarios. In this paper, we pay particular attention to air pollution since emissions from energy production and utilization contribute to the local air quality significantly. Moreover, the mitigation of air pollutants has become a major concern in climate action [37]. This paper considers the mitigation of PM₁₀, NO_x, SO₂, and NMVOC due to the alternative low carbon development pathways.

Sensitivity analysis for the household sector

To improve our understanding, sensitivity analysis for the household sector of Nigeria was conducted. The household sector accounts for around 80% of the total final energy consumption in the country as per the IEA energy balances of Nigeria, 2015 [38]. This figure is very high and, thus, analysis of energy demand reduction potentials of the sector will be useful. The sensitivity analysis was conducted to quantitatively assess the impact of each residential sector lever on the overall energy demand of the country. In doing so, the residential sector levers were changed from one (no efforts) to two (little efforts); two to three (greater efforts); and then three to four (maximum efforts), while keeping other sectors at reference scenario (S1) levels. This was done to allow for comparison with respect to the reference scenario. Moreover, this particular analysis provides insights into the scale of energy demand reductions through demand-side management techniques in the household sector. It is worth noting that the sensitivity analysis of the household sector does not overcome the inherent uncertainties of the model such as the aggregation of key sectors and the projections of economic indices (e.g. GDP growth and fossil fuel prices).

Results

Final energy demand

Figure 3 shows the final energy demand for S1–S5 scenarios for the period 2010–2050 based on the macroeconomic, technological, and behavioral assumptions used in the calibration of NECAL2050. Results show that energy demand will increase steadily across the five scenarios considered in this study as a result of the increase in population as well as socio-economic activities. However, it is observed that the average annual growth rate varies across the scenarios. In the S1 scenario, final energy

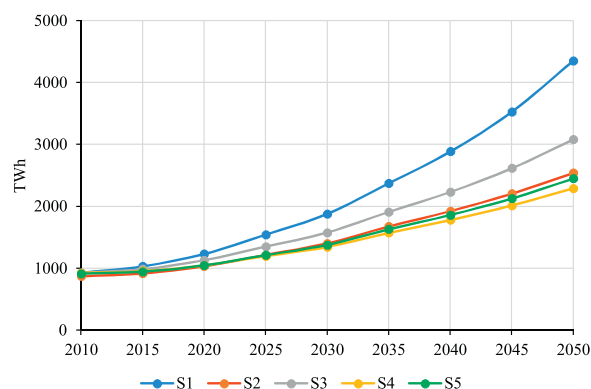
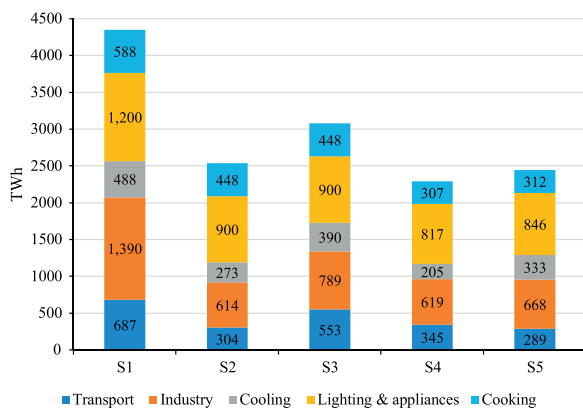


FIGURE 3
Final energy demand by scenarios.

**FIGURE 4**

Sectoral final energy demand in 2050.

demand is expected to grow at an average annual rate of 3.14% from 926 TWh in 2010 to 4353 TWh in 2050. This is around four times increase compared to 2010 value and the highest among all other scenarios. Our analysis (See Figure 4) indicates that by 2050, industry will consume the highest amount of energy with a share of around 40%, followed by lighting & appliances with 27.6%. Transport, cooking, and cooling are seen to contribute 15.8%, 13.5%, and 11.2%, respectively, to the final energy demand by 2050. In S2, with the introduction of energy-efficient appliances in the industry, commercial and households sector as well as the introduction of energy-efficient vehicles in the transport sector, final energy demand is observed to rise from 870 TWh in 2010–2539 TWh in 2050 at an annual growth rate of 2.17%. This implies a 41.7% reduction in energy demand or 1814 TWh relative to the S1 scenario by 2050. In the same year (Figure 4), the final energy demand for lighting & appliances was the highest with a share of 35.4%. Significant reductions are also observed in the industry, cooling, and cooking sectors. It is important to note how radical these changes are across the various scenarios.

In the S3 scenario, energy-efficient appliance penetration was low and final energy demand grew from 926 TWh in 2010 to 3080 TWh in 2050 at an annual rate of 2.43%. When compared to the S1 and S2 scenarios, analysis shows that energy demand will fall and rise by 29.2% (1273 TWh) and 21.3% (541 TWh), respectively. Sectoral analysis of the S3 scenario shows that lighting & appliances will account for 29.2% which is the highest sectoral energy demand, followed by the industry sector having a share of 25.6%. Final energy demand for lighting & appliances and cooking (900 TWh and 448 TWh, respectively) are the same in the S2 and S3 scenarios. With the high penetration of bioenergy supply in the S4 scenario, final energy demand will rise with an annual growth rate of 1.29% from 926 TWh in 2010 to 2293 TWh by 2050. This is equivalent to 47.3%, 9.7%, and 25.6% reductions when compared with S1–S3 scenarios, respectively. S4 scenario generates the highest reduction in energy demand among the five scenarios considered in this paper. At sectoral levels, though, lighting & cooking accounted for the highest percentage of energy demand (35.6%), the value (817 TWh) is lower compared to other scenarios. Our analysis also shows that final energy demand in the cooling and cooking sectors of S4 scenario are the lowest among the remaining four scenarios.

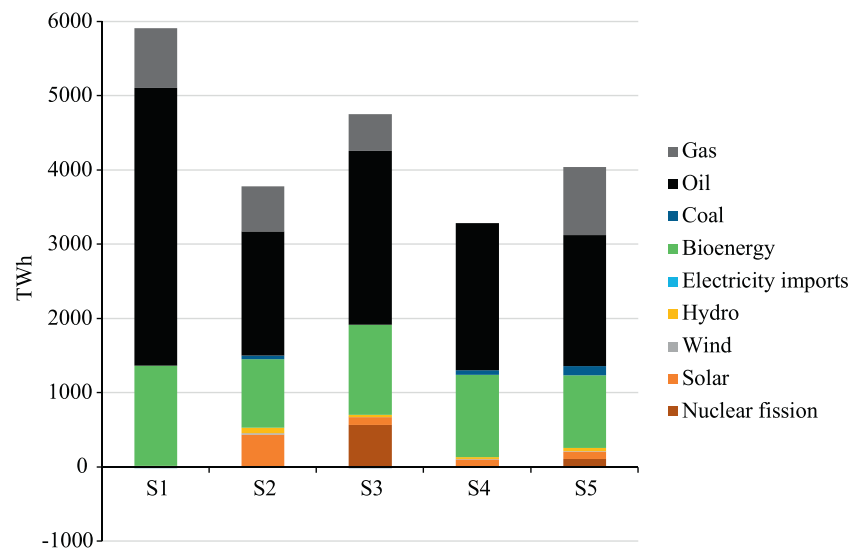
In S5 scenario, efforts towards reducing energy consumption as well as GHG emissions are distributed across all sectors of the economy. Final energy demand will rise at an annual rate of 2% from 912 TWh in 2010 to 2449 TWh in 2050. When compared to the S1–S3 scenarios, by 2050, final energy demand will fall by 43.7%, 3.5%, and 20.5% (or 1904, 90, and 631 TWh), respectively. While recording these reductions, we observed that a similar trend was not applicable to the S4 scenario. Our analysis for S5 scenario shows that final energy demand will increase in 2050 by 6.8% or 156 TWh relative to S4 scenario.

In sum, our results suggest that lighting & appliances and industry sectors will consume the highest amount of energy, accounting for over 60% of final energy demand by 2050 in all scenarios. The projected increase in demand for lighting & appliances in the country may be attributed to the projected increase in population and households' income, as well as the general yearning for socio-economic development in the country. Nigeria also has a vision 20:2020 target—to become one of the twenty biggest economies in the world by 2020 and beyond. Whether this target will be achieved or not is beyond the realm of this paper. However, the industrial sector will play a major role towards this target and may be the reason for the projected increase in the industrial sector energy demand.

Final energy supply by source

From Figure 5, it is observed that final energy supply increases across all scenarios by 2050, as is the case with final energy demand. In S1 scenario, final energy supply by 2050 is calculated as 5902 TWh. Final energy supply in the S2–S5 scenarios are observed to be 3778, 4738, 3287, and 4039 TWh, respectively which corresponds to some reductions of 36%, 19.7%, 44.3%, and 31.6%, respectively, when compared with the S1 scenario. With respect to the sources of energy supply in S1 scenario by 2050, fossil fuels will dominate the Nigeria energy supply system with oil and gas accounting for around 63.4% and 13.7%, respectively. On the renewables side, bioenergy which is mainly used for domestic cooking continues to be the most prominent renewable energy consumed in the country and accounts for over 90% of renewable energy supply in the country as well as 23% of the final energy supply system. The share of hydro and solar are seen to be less than 1% of the total energy supply, while nuclear is kept out of the system. However, with the deployment of large-scale renewable energy technologies in the S2 scenario, we observed that the share of fossil fuels in the energy supply system reduced when compared with S1 scenario and oil accounted for around 44.2%. With respect to renewables, results show that solar, wind, hydro and bioenergy will account for about 11.5%, 0.6%, 2%, and 24.3%, respectively in the energy supply system, while nuclear energy is still left out by 2050.

In S3 scenario, nuclear energy was introduced with moderate penetration of renewables and our analysis show that oil will dominate the supply system with a share of 49.4%, while gas will have a share of 10.5%. Nuclear energy is seen to account for 13% of the supply system whereas solar, wind, hydro and bioenergy will account for 2.2%, <1%, <1%, and 25.5%, respectively. In S4 scenario, the supply of bioenergy was enhanced for modern bio-fuels production, consumption of fuelwood reduced in the residential sector, other renewables were also employed moderately

**FIGURE 5**

Final energy supply in 2050.

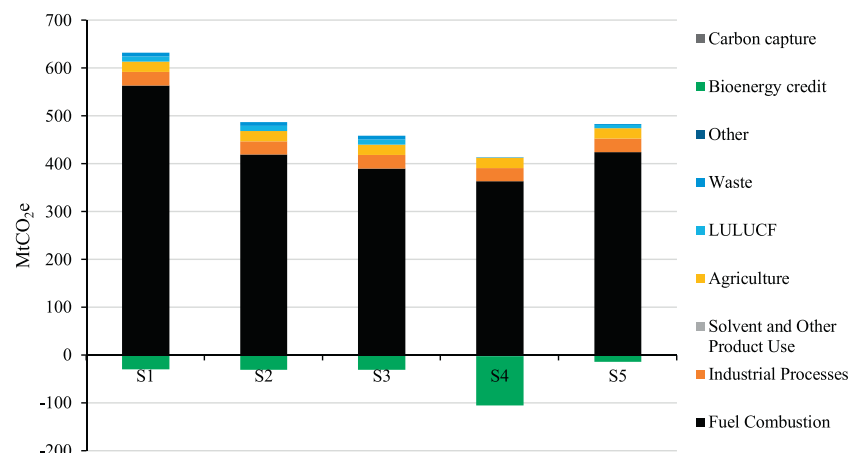
and nuclear was kept at the reference scenario level. Results show that oil will account for around 60% of the total supply system whereas bioenergy will account for 33.6%. Solar contributes around 3.2% to the primary energy supply while other renewables are seen to account for just about 1% of the total supply system. Analysis for the S5 scenario shows that fossil fuels will have a greater percentage in the supply system compared to renewables and nuclear. In general, oil will have the greatest percentage of the Nigeria energy supply system in all scenarios by 2050.

GHG emissions

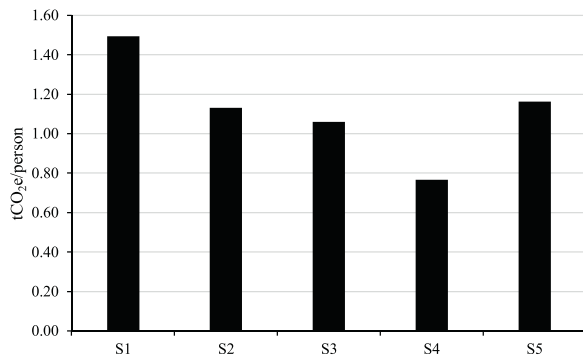
Total GHG emissions for the five scenarios in 2050 are shown in Figure 6. We observe that GHG emissions from the S1 scenario are the highest at around 604 MtCO₂e. With the introduction of low carbon technologies such as renewable and nuclear energy technologies, GHG emissions are seen to fall by 24.3%, 29%, 48.8%, and 22.2% in the S2–S5 scenarios respectively as compared with the S1 scenario. We observe that maximum GHG emission reduction was recorded in the S4 scenario. This can be attributed to the

high bioenergy and renewables penetration in that scenario which are both net zero carbon emitters. Our analysis indicates that the lowest GHG emission will be recorded in S5 scenario. It may be recalled (Figure 3) that S5 scenario produced the second lowest energy demand, however, since fossil fuels still account for a very significant portion of the energy supply system in this scenario, GHG emission is expected to rise when compared to S2–S4 scenarios.

With respect to sources of GHG emissions, fossil fuel combustion is seen to account for the highest GHG emissions (over 80%) across the five scenarios. This is followed by industrial processes and then agriculture for all the scenarios. Land use, land-use change, and forestry (LULUCF) is also seen to contribute around 1% to total GHG emissions across all scenarios in exception of S4 which is as a result of high bioenergy penetration in that scenario. The same trend was also seen in bioenergy credit with S4 scenario having the highest bioenergy credit at around 103 MtCO₂e. This is followed by S2, S3, S1, and S5 scenarios having around 30.6, 30.5, 28.8, and 13.7 MtCO₂e respectively. Though carbon capture and

**FIGURE 6**

GHG emissions by sources in 2050.

**FIGURE 7**

Per capita GHG emissions by 2050.

storage (CCS) was outlined in the NECAL2050, it has not been considered here since the model developing team didn't consider it as an option for Nigeria by 2050.

Sometimes, per capita GHG emissions is used to assess the GHG emission intensity of a country as well as total national GHG inventory. It also provides the framework for comparing national GHG emissions among countries across the world. Figure 7 shows the per capita GHG emission for Nigeria under the five scenarios in 2050. As expected, S1 and S4 scenarios have the highest and lowest per capita GHG emissions with 1.50 and 0.77 tCO₂e/person, respectively.

Electricity demand

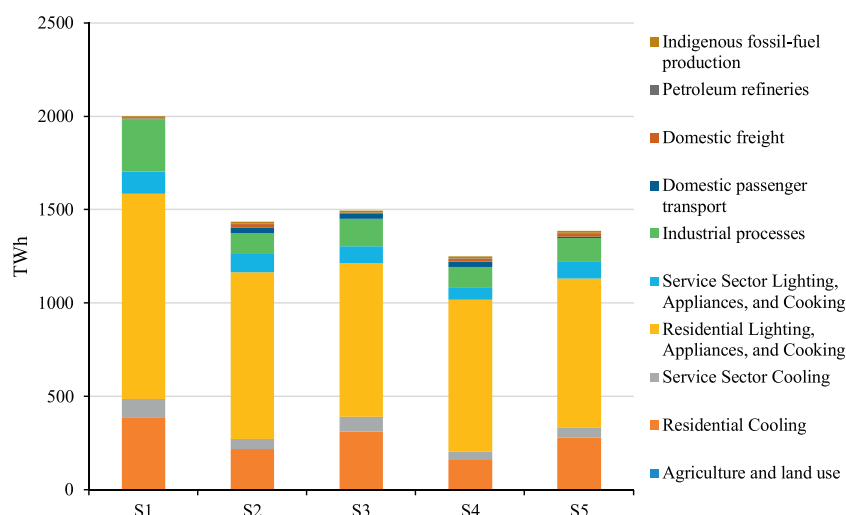
Given the importance of electricity in this era, electricity consumption is analyzed separately and is shown in Figure 8. By 2050, total electricity consumption in the S1 scenario is seen to be 2004 TWh. Results suggest that electricity demand will drop significantly in the remaining four scenarios when compared with S1. Our analysis shows that by 2050, electricity demand will drop by 28.3%, 25.3%, 37.5%, and 30.8% or 567, 507, 753, and 617 TWh in the S2–S5 scenarios, respectively. It may be noted that S3 and S4 scenarios have recorded the lowest and highest electricity demand reductions, respectively, relative to S1 scenario. The lower electricity demand reduction recorded in S3 may be attributed to the

penetration of low energy-efficient appliances in that scenario which in turn consumes more amount of energy.

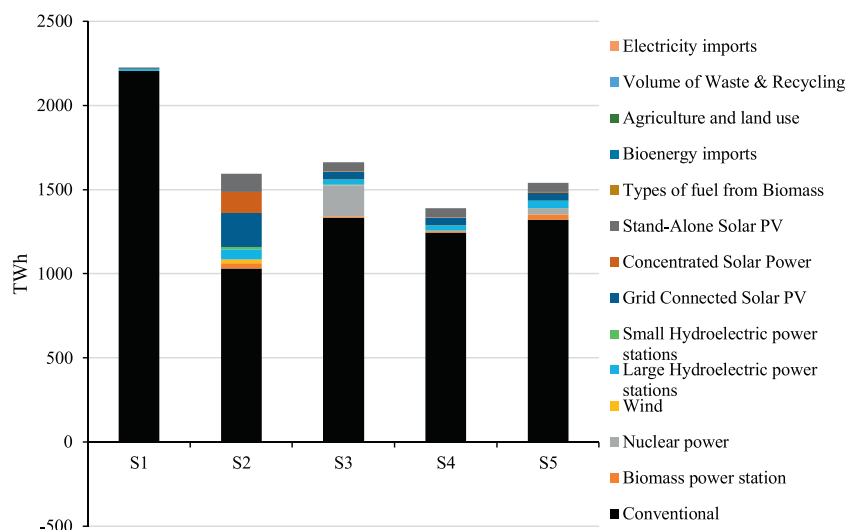
At sectoral levels, residential lighting, appliances, and cooking sector consume over 50% of electricity under the five scenarios. This is followed by residential cooling and then industrial processes accounting for around 20% and 8.7%, respectively, across all the scenarios. The domestic passenger transport sector contributes around 2.2% in the final electricity demand in S2–S4 scenarios. However, when efforts are shared in the S5 scenario, domestic passenger transport accounts for <1% of final electricity consumption. Agriculture is seen to have the lowest electricity consumption and accounts for <1% of the total electricity consumption in all scenarios. This may be attributed to the historically low level of electricity consumption in the sector as well as the use of manual power in the sector. Electricity consumption of petroleum refineries and indigenous fossil-fuel production were observed to be 3 and 10 TWh in all the scenarios. Graphical illustration of these results is presented in Figure 8.

Electricity supply

Figure 9 shows the results of final electricity supply by 2050 under the five scenarios. To constantly match electricity supply to demand in the S1 scenario, around 2226 TWh will be generated with the conventional sources (gas, coal, and oil) accounting for 99% of the final electricity supply, which indicates no efforts towards limiting GHG emissions in the country. However, in the alternative scenarios, total electricity supply is seen to drop significantly compared to the S1 scenario. S2 scenario has a final electricity supply of 1596 TWh. With the introduction of renewable energy technologies, the conventional source is still seen to account for around 65% of the total electricity supply. In the same scenario, biomass power, wind, large hydro, small hydro, grid-connected solar PV, concentrated solar power, and stand-alone solar PV account for about 2%, 1.4%, 3.7%, 1%, 12.6%, 7.7% and 6.9%, respectively, in the final electricity supply. In the S3 scenario, electricity supply will be 1663 TWh by 2050. Nuclear energy is introduced aggressively into the Nigerian energy mix whereas renewables are kept at a moderate level.

**FIGURE 8**

Final electricity demand in 2050.

**FIGURE 9**

Final electricity supply in 2050.

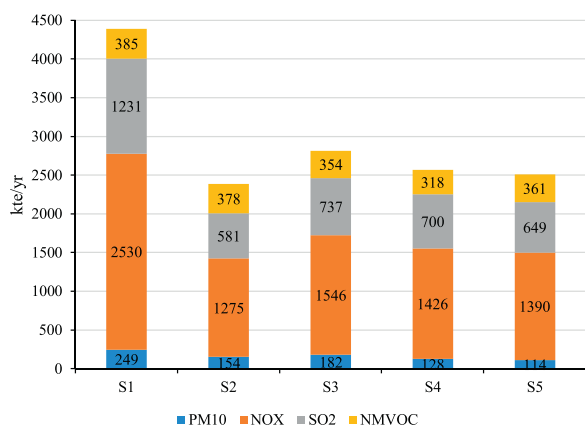
This analysis suggests that conventional sources and nuclear energy will account for around 80% and 11% of the total electricity supply, respectively. Stand-alone and grid-connected solar PV are seen to contribute 3.3% and 2.8%, respectively. Large hydro accounts for 1.7% while biomass, wind, concentrated solar power, and small hydro each account for <1% of the total electricity supply. In S4 scenario, total electricity supply will be 1390 TWh in 2050. Conventional carbon-intensive energy will dominate the electricity supply system with a share of about 89%. When efforts are distributed in the S5 scenario, total electricity supply will be 1541 TWh. This analysis suggests that conventional sources of electricity will dominate the supply system with around 85.8%. On the other hand, biomass, nuclear, wind, large hydro, small hydro, grid-connected solar PV, concentrated solar power, and stand-alone solar PV will account for 2.1%, 2.3%, <1%, 2.8%, <1%, 3%, <1%, and 3.6%, respectively.

Air pollution and sensitivity analysis results

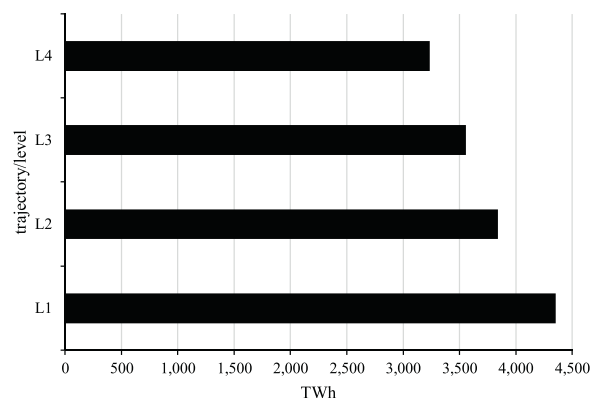
Results of air pollutant emissions by 2050 for all the scenarios considered here are presented in Figure 10, in kilotons equivalent of air pollutant per year (kte/yr). It may be seen that emission of air

pollutants decreases in all the alternative scenarios when compared to the reference case. The same trend is observed for all types of air pollutants. However, it should be noted that air pollutant emissions vary across the scenarios. For instance, in the alternative scenarios, NO_x emissions are lowest in S2 whereas NMVOC emissions are highest in the same scenario. These variations of air pollutants emissions across the different scenarios should, therefore, be considered while understanding the implications of these scenarios. All in all, the general reduction of air pollutant emissions should be seen as important benefits of low carbon development as the health concerns associated with air pollutants like these ones are very problematic. Thus, sustainable energy transition would provide greater air quality and the benefits that would accrue from the avoidance of these air pollutants should be factored in when making low carbon development policies.

Sensitivity analysis for household sector shows that total final energy demand in the country will drop from 4353 TWh in level (L) 1 or reference case to 3838, 3554, and 3233 TWh in L2–L4 trajectories, respectively. The graphical results are shown in Figure 11. The result indicates that with the massive introduction of energy-efficient appliances in the household sector, the sector

**FIGURE 10**

Air pollutant emissions in 2050.

**FIGURE 11**

Household sensitivity analysis of final energy demand.

has the potential to reduce the country's overall energy demand in 2050 by around 25.7% relative to the reference case. Thus, energy efficiency practices should be the key focus of policymakers with respect to the residential sector.

Discussion and policy implications

Realising a low carbon Nigeria by 2050 depends on the views of the government and the scale of efforts towards the goal. Across the four alternative pathways considered in this study, we observe that energy demand and GHG emissions will reduce relative to the business-as-usual case. Our analysis indicates that S4 scenario will have the highest amount of GHG emission reduction but with the technical limit of bioenergy supply in the country. However, for effective realization of a low carbon Nigeria and energy security by 2050, there is a need for policy trade-offs between energy demand reduction and GHG emissions mitigation. Efforts towards energy demand reduction should be focussed on the demand side while the supply side should be the focus of limiting GHG emissions. Achieving the technical limit of any of the trajectories across all sectors will require huge dedication by the Nigerian government and citizens.

Thus, emphasis needs to be placed on the specific measures that would keep Nigeria on the track of low carbon transition by 2050. On the demand side, the transport sector provides options for energy demand reduction and GHG mitigation. This can be achieved through the combination of technological and behavioral changes. On the technical side, the introduction of energy-efficient vehicles that run on biofuels, electricity, compressed natural gas (CNG) and fuel cells are needed in the Nigerian transport sector. However, this may not be achieved so fast since about 60% of Nigerians still live below the poverty line and may not afford such kinds of vehicles. Also, with the present electricity crisis in the country, the use of electric vehicles may not be feasible in the near future as it has to compete with other sectors for electricity supply. Thus, efforts should be made towards infrastructure development in the country, while those at the upper-income class can start the purchase of energy-efficient vehicles. Moreover, the government can play a role here by providing subsidies for energy-efficient vehicles as well as local manufacturing which will, in turn, make the vehicles affordable. With respect to behavioral changes, the main objectives are to shift mode of transport from cars to buses, increase vehicle occupancy ratio, and telecommuting. This will, however, require the introduction of taxes on car owners and also the provision of huge incentives for bus/lorry operators. These actions would significantly reduce the demand for transport and, consequently, reduce the demand for energy.

The household and service sectors also provide opportunities for low carbon transition. The main concern here should be the total phase-out of incandescent bulbs and the introduction of efficient cookstoves. Other measures like the introduction of minimum energy performance standards for other appliances such as television, air conditioners, and refrigerators should be considered. Furthermore, renewable energy technologies such as rooftop solar PV will go a long way to reduce the carbon footprint of the Nigerian residential sector. An earlier study has argued in the favor of this assertion and opine that rooftop solar PV is one of the viable options for energy security and climate mitigation in Nigeria [39]. The industrial sector is also not left out in the journey towards low-carbon development. Emphasis should be laid on the

use of energy-efficient equipment in this sector. Carbon capture and sequestration (CCS) offers another option for GHG mitigation. However, there are many uncertainties regarding this option as it is very expensive to implement and has not been considered as a policy lever even in NECAL2050. Though options are offered for limiting energy demand and GHG emissions in the various sectors of Nigerian economy, it is important to mention that the low carbon pathways explored in this study will be very challenging to attain as it will require all-hands-on-deck including the government and civil societies. The pathways explored shows the level of efforts required by each actor towards the realization of Nigeria's low carbon development strategies.

The S2 scenario (High renewables + high energy efficiency + behavior) would require great efforts by the government to decarbonize the electricity sector by deploying large-scale renewable energy technologies. This further implies that the present electricity grid in the country needs to be expanded and fully refurbished since it is characterized by high rate of losses [40]. Also, for effective management of the electricity sector, the Nigeria government need to consider the deployment of smart grid technology in the country. It may also be noted that renewable energy supply is intermittent in nature and thus, the use of energy storage technologies may be required in the country. Technologies like molten salt, pumped hydro, and battery may be deployed in the Nigerian energy system. For investors to have the propensity and confidence to invest in renewable energy technologies in Nigeria, the government need to also provide a robust feed-in tariff scheme in order to ensure maximum returns for investors. S2 scenario also suggests a high energy efficiency practice as well as behavioral changes. This will mainly affect the consumers or the Nigerian citizens in this case. This pathway will involve trading short economic benefits for good quality of life in the future. For example, the initial capital outlay for a diesel generator set is small when compared with a solar home system. However, in the long run, a solar home system will appear to be a cheaper means of electricity generation. Hence, for consumers to imbibe this sustainable lifestyle, there is a need for massive education on the benefits of sustainable development across the country while attention should not be taken away from cultural preferences as in the case of cooking equipment.

The S3 scenario (high nuclear + moderate renewables + low energy efficiency) suggested the introduction of large nuclear power facilities in the country. Here, the main concern is how Nigeria can manage the technology. At present, there is no nuclear facility in the country and the government does not appear ready to install nuclear power in the near future. Thus, the first task should be identifying strategies for management of nuclear waste as well as disaster management in case of any incidents in the future. In addition, investments should be made towards building manpower in that area. At present, only a handful of Nigerians are skilled at nuclear technology.

Given the abundance of bioenergy in the country, the S4 scenario (High bioenergy + high energy efficiency + moderate renewables) assumes that the technical limit of bioenergy in the country is reached (as per NECAL2050 design) and will produce the highest GHG emission reduction among other scenarios. However, sustainability of biomass remains an issue. Bioenergy is already part of the energy supply system of the country especially in the household sector and will likely play a significant role in the low carbon

transition. The Nigerian government needs to tap the full potential of bioenergy resources in the country before seeking for importation if the S3 scenario is to be followed. However, it is worth mentioning that while this scenario is attractive from the perspective of GHG emissions mitigation, it might impact the natural ecosystem as well as those that depend on biomass for livelihood. Hence, sustainability assessment is first step required while working on this scenario.

In S5 (shared efforts), significant energy demand reduction will be recorded but with minimal GHG emissions reduction. Here, efforts are shared equally but only one objective is achieved. This scenario may be the easiest to achieve in Nigeria since none of the sectoral technical limits are reached. This implies that for Nigeria to significantly reduce its current GHG emissions by 2050, it needs to reach its technical limit in at least one of the supply sectors; although this may be very challenging, it is attainable. In addition, efforts towards a low carbon Nigeria could open up some economic opportunities for Nigeria. In recent times, the cost of renewable energy technologies has been on the decline and may continue in the same trend in the future. Thus, Nigeria can exploit this opportunity to decarbonize its power sector at minimal cost. This will also open new markets in the Nigerian energy sector that can employ a significant amount of the teeming unemployed youths in the country. Our analysis also shows that low carbon transition will improve the air quality and consequently improve the health condition of many Nigerians. In the cooking sector, the shift from traditional solid fuels to clean and modern cooking equipment will reduce the amount of deaths attributed to indoor air pollutants in the country, which will, in turn, save the money that would have been spent on health care. This would also allow householders, especially those in rural areas, to engage in income-producing activities with their saved time as they would no longer have to spend hours gathering fuelwood from the forest. In a nutshell, if the Nigerian government does not pay serious attention to the foregoing concerns, then the realization of a low carbon Nigeria will most likely remain a fantasy and may never become a reality by 2050.

Conclusions

The Nigerian energy system is presently carbon-constrained and it is crucial to have an idea on the scale of efforts required towards decarbonizing the country. Many of the development choices the country makes today will determine its carbon footprints by 2050. This paper has explored four alternative low carbon development pathways for the country. The reference scenario (S1) which follows the business-as-usual or conventional development pathway shows little concern for low carbon transition in the development strategy of the country. However, the alternative scenarios (S2–S5) consider low carbon transition as an issue to be included in the broader development plan of the country. Nigeria's government proposes to achieve its developmental goals in a sustainable manner. We find that the alternative scenarios show some reductions in energy consumption, GHG emissions, and air pollution when compared to the reference case, but fossil fuels will still dominate the Nigerian energy system in all scenarios by 2050. Thus, efforts that successfully decouple economic development from GHG emissions are still needed to keep Nigeria on the right path of low carbon transition. Based on our modeling assessments, we may conclude that the current sustainable energy policies in the country will certainly reduce GHG emissions and improve air quality but will not

decarbonize the country significantly by 2050. We, therefore, recommend that low carbon policies should be targeted at the key sources of GHG emission in the country viz. fossil fuel combustion. Specifically, efforts which combine the full exploitation of bioenergy and energy efficiency practices (S4 scenario) will have the maximum potential for GHG mitigation in both supply and demand side of the Nigerian energy system. While other renewables such as solar and wind are needed in the low carbon transition 'story' of Nigeria, we suggest aggressive deployment of biofuels in the Nigerian energy system. Particularly, we suggest fuel switching towards biofuels such as biodiesel and bioethanol for the road transport sector of Nigeria which presently runs fully on diesel and gasoline.

Furthermore, the Energy Commission of Nigeria (ECN) has started an energy efficiency campaign in conjunction with the United Nations Development Programme (UNDP) and the Global Environment Facility (GEF). However, this campaign needs to be enhanced, especially in the residential and industrial sector. For examples, the programme needs to include Energy Performance Index (EPI) for buildings, enhanced recycling of materials such as scrap-based steel, use of fly ash and slag in the cement industry as well as the use of halocarbons with low global warming potentials for refrigerants. While these policies can spur Nigeria on the path of low carbon transition, it is worthwhile to note that investments in low carbon technologies will be a key factor for policymakers while considering the aforementioned recommendations. Currently, there is no coherent funding for low carbon development projects. We suggest the establishment of a low carbon development fund in the country. Achieving this within the country will be particularly helpful for seeking external funds from international donors and bilateral organizations such as the African Development Bank, the World Bank as well as the Green Climate Fund (GCF) of the Paris agreement, in order to augment the shortfalls in local funds. Finally, it is worth mentioning that there are some caveats in our analysis. The 2050 Calculator modeling framework is not flexible compared to other energy system modeling tools such as LEAP and MARKAL/TIMES. Hence, a tolerance of $\pm 5\%$ should be taken into consideration while using the results of our analysis. Additionally, NECAL2050 does not give the macroeconomic implications of each low carbon pathway. While noting these limitations, we believe that the results of our analysis will support policy decisions with respect to low carbon transition in Nigeria.

Conflicts of interest

The authors have no conflict of interests to declare.

Acknowledgments

We thank Prof. Eli Jidere Bala, the Director General of the Energy Commission of Nigeria for granting the permission to use the NECAL2050 model.

Appendix A NECAL2050 development

NECAL2050 was developed based on the general model framework of the UK 2050 Pathways Calculator. The main purpose for developing this tool was to provide a platform for experts and non-experts in the energy domain to develop their own future energy and GHG scenarios based on different technological combinations. To adapt

the model to the Nigerian condition, several changes were made in it in order to reflect the actual energy situation of Nigeria. The fundamental changes that were made are as follows:

- Extensive literature review as well as experts' consultations were made to gather Nigeria-specific data on macroeconomic variables, technologies characterization, and scenarios setting.
- Economic sectors were set to portray the key energy demand and supply sectors in Nigeria.
- Macroeconomic and social indicators for 2050 for four scenarios were set at the upper level of the model framework and its been linked with the important sectoral scenarios for both the demand and supply sides.
- With respect to the supply side, four trajectories were set to reflect the extent of efforts on energy resources or technological penetration in the supply sector viz; (i) no efforts (ii) medium efforts (iii) high efforts, and (iv) extreme efforts.
- Nigeria situations were considered while providing technological options, as some technologies available in the UK Pathway Calculator were not considered feasible in Nigeria.
- In NECAL2050, energy supply has been set to constantly match demand by balancing the gap between electricity supply from the grid and demand with self-electricity generation by generators.
- Levels of cost were not included in the current version of NECAL2050.

Structure of NECAL2050

In this sub-section, we report the basic structure of the model. NECAL2050 is structured into three different components which are (i) energy demand (ii) energy supply, and (iii) non-energy sectors.

(A1) Energy demand component

The energy demand sectors covered are residential, service, transport, and Industry.

(A1) Residential and service sectors

Energy services considered in the residential and service sectors includes cooking, lighting & appliances, and space cooling. Framework for analysing energy demand in these sectors are given as follows:

lighting and electrical appliances (LD_e) are calculated as:

$$LD_e = LD_e / H \times H_n \quad (A1)$$

Where LD_e / H accounts for the electricity demand for lighting and electrical appliances per building, while H_n is the number of building in the residential/service sector.

Cooking energy demand by fuel type in the residential or service sectors is calculated as:

$$CKD_f = CKD_f / H \times H_n \quad (A2)$$

Where CKD_f / H accounts for the total energy demand by fuel type for cooking in a building, while H_n is the number of building in the residential or commercial or service sector.

The total energy demand in the residential or service sector i at time t is:

$$TD_{it} = LD_e + CKD_f \quad (A3)$$

While the associated greenhouse gas emission (GHG_{it}) is calculated as:

$$GHG_{it} = TD_{it} \times E_{fct} \quad (A4)$$

Where E_{fct} accounts for the emission factor of the fuel type at time t .

The space cooling energy demand which in this case is electricity is calculated as:

$$SC_e = SC_e / H \times H_n \quad (A5)$$

Where SC_e / H accounts for the electricity demand for space cooling per building, while H_n is the number of building in the residential or service sector.

• Transport sector

This sector consists of seven different modes of transport and are presented here with their associated technologies:

- Air: aeroplane
- Bike: bike
- Boat: ferry
- Bus: internal combustion engine (liquid and gaseous hydrocarbon); electric vehicle (EV), and hybrid electric vehicle (HEV)
- Car: internal combustion engine (liquid and gaseous hydrocarbon); electric vehicle (EV), and hybrid electric vehicle (HEV)
- Motorcycle and tricycle
- Train: diesel and electric

Total passenger-km travel by mode is calculated as:

$$MS_t = DT_m / TDT \times 100 \quad (A6)$$

Where MS_t is the mode share at time t , DT_m is the distance travelled mode, while TDT is the total distance travelled which is given as:

$$TDT = DT_{pt} \times Pop_t \quad (A7)$$

Where DT_{pt} is the distance travel per person at time t . Pop_t is the population at time t .

The passenger-km by mode or technology is calculated as:

$$O_t = V_n / Pop_t \quad (A8)$$

Where O_t denotes the ownership of vehicles at time t , V_n is the number of vehicles at time t , and Pop_t is the population at time t .

The occupancy (OC_t) and technology penetration are calculated as shown in Eqs. (A9) and (A10) below:

$$OC_t = P_{nt} / V_t \quad (A9)$$

$$TC_t = Tec_t / \sum T_m \times 100 \quad (A10)$$

Where P_{nt} is the number of persons at time t , V_t is the vehicle at time t , Tec_t is the technology at time t , and $\sum T_m$ is the technology used in the transport mode at time t .

Energy demand (D_t) by mode and fuel type for passenger and freight transport are calculated as:

$$D_t = \sum DT_m + DT_T / I_T \quad (A11)$$

Where $\sum DT_m + DT_T$ is the summation of the distance travelled by mode and technology, while I_T is the intensity or efficiency of

the technology which is calculated as:

$$I_T = F_l / DT_{km} \quad (A12)$$

Where F_l is the fuel used per litre and DT_{km} is the distance travelled in km.

The greenhouse gas emission by mode of transport is calculated as:

$$E_{GHG} = \sum DT_T \times E_{fct} \quad (A13)$$

Where $\sum DT_T$ is the summation of distance travelled by technology and E_{fct} accounts for the emission factor of the fuel type at time t which is calculated as:

$$E_{fct} = CO_{2kg} / DT_{km} \quad (A14)$$

Where CO_{2kg} measures the kilogram of carbon emission equivalent.

(A15) Industry sector

The energy demand in the industry sector is computed as the sum of energy demand from the key industrial sub-sectors (construction, manufacturing, and mining). For the industry sector, the total energy demand (TED) is given as:

$$TED = TD_{man} + TD_{con} + TD_{min} \quad (A15)$$

Where TD_{man} , TD_{con} and TD_{min} represent energy demand from the manufacturing, construction and mining industries respectively.

The energy demand by industry subsector (D_{sub}) is given by:

$$D_{sub} = OI \times M \quad (A16)$$

Where OI is the output index of the industry subsector, while M is the energy demand multiplier.

The energy demand by fuel is:

$$D_f = TED \times F \quad (A17)$$

Where F is the share of fuel in the fuel mix.

The process emission for given greenhouse gas (P_{GHG}) is:

$$P_{GHG} = P_{GT} \times N \quad (A18)$$

Where P_{GT} is the process emission growth and N is the baseline greenhouse gas emission. The process emission (P_{GHG}) are emissions from industrial processes which includes CO_2 , CH_4 , N_2O , HFCs, PFCs, SF6, PM10, NOx, SO_2 , and NMVOC.

The combustion emission (C_{GHG}) is:

$$C_{GHG} = D_f \times E_{fct} \quad (A19)$$

The combustion emission (C_{GHG}) are gases emitted during fuel combustion for the generation of energy required for industrial processes.

(A20) Energy supply component

Primary energy supply resources in NECAL2050 include coal, crude oil, natural gas, nuclear, solar, wind, and biomass. Production bounds has been placed for primary energy resources in the model for a central, lower, and very low production cases. Details of these bounds and other resources bounds are written in the model which is provided here in Appendix B. Secondary energy supply encompasses electricity, biofuels, and oil products. The electricity supply technologies include non-renewable and renewable energy technologies. The non-renewable energy technologies

are: generators set (gasoline and diesel), gas power plant, coal power plant, nuclear power plant, and biomass power plant. The renewable energy technologies are wind turbines, solar power plants (PV & CSP), and hydro power plants (large & small). Energy supply calculations are presented as follows:

$$AG = AC \times LF \quad (A20)$$

Where AG is the available generation, AC the available capacity, and LF the load factor.

Thus, total generation (TG) will be:

$$TG = AG \times PoR \quad (A21)$$

Where PoR is the plant's own requirement.

The resources requirement RQ_t at time t is calculated as:

$$RQ_t = T_G / I_{TH} \quad (A22)$$

Where I_{TH} is the thermal efficiency of the power plant.

For fossil-based power plants, GHG emissions ($GHGe$) is computed as:

$$GHGe = RQ_t \times EF \quad (A23)$$

Where EF is the emission factor of the fuel consumed by the power plant.

Non-energy component

This component basically consists of agriculture and land use, types of fuel from biomass, as well as waste and recycle. The land use component was included in the model to capture the area of land that will be dedicated to bioenergy in any scenario. Around 50% of Nigeria renewable energy resource potentials are from agricultural and forest residues. The landmass of Nigeria is around 923,768 square kilometers. The fuels from biomass considered in the model are biodiesel, bioethanol, and biogas. Owing to increase in population, urbanization, and living standards, its is expected that municipal solid wastes (MSW) will increase in the near future. Thus, MSW was included in the model as alternative fuel for waste-to-energy technologies.

Appendix B

NECAL2050 model (<https://data.mendeley.com/datasets/35b6b33jpb/draft?a=5a8efa62-d8d3-4a55-ba3a-ffaa991be7f4>).

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